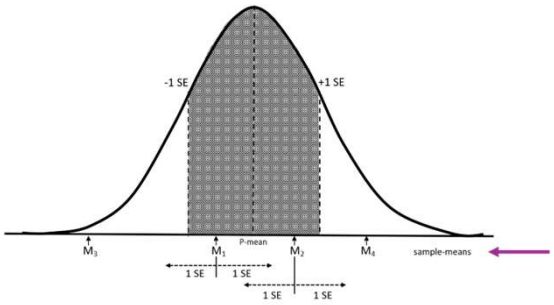
Agenda

- Last time
  - χ^2_n distribution
  - The S^2 statistic
  - Independence of S^2 and $\bar{X}$
- Today
  - Confidence interval



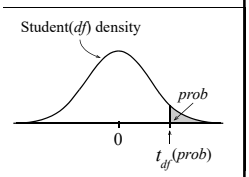
1

Approximate confidence interval

- Earlier: Cavendish conducted experiments seeking density of earth
  - sample mean of 29 observations $\bar{x}$ = 5.447931 g/cm^3 and s_x = 0.2209457 g/cm^3 and therefore se($\bar{x}$) = 0.04102858
  - True mean lies in the interval $\bar{x} = 5.447931 \pm 2 \times 0.04102858$
- Assumptions and problems
  - Experiments may be "biased"
  - Distribution of means is only approximately normal
  - Too many significant digits
- Coarse bound can be strengthened with the "student t-distribution"

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Confidence interval with t-distribution



- Earlier: Cavendish conducted experiments with 29 values
  - $\bar{x}$ = 5.447931 s_x = 0.2209457 se($\bar{x}$) = 0.04102858
  - True mean lies in the interval $\bar{x} = 5.447931 \pm 2 \times 0.04102858$
- Coarse bound can be strengthened with the "student t-distribution"
  - $(\bar{X} - \mu)/(S/\sqrt{n}) \sim t_{n-1}$
  - $\Pr\left(-t_{0.025,n-1} \leq \frac{\bar{X}-\mu}{S/\sqrt{n}} \leq t_{0.025,n-1}\right) = 0.95,$
  - tinv(0.975,28) = 2.0484, so $\bar{x} \approx 5.45$, confidence interval = [5.37, 5.53]

Note 28, not 29

Typo 5.36

4

Significant digits

- Cavendish conducted experiments with 29 values
  - $\bar{x}$ = 5.447931 s_x = 0.2209457 $\mu \in 5.447931 \pm 2.0484 \times 0.04102858$
- Coarse bound can be strengthened with the "student t-distribution"
  - tinv(28, 0.975) = 2.0484, so $\bar{x} \approx 5.45$, confidence interval = [5.36, 5.53]
- Rules of thumb for significant digits
  - Intermediate calculations as accurate as possible
  - Estimate to be reported up to the first significant digit of se, se one more significant digit than the se, confidence interval maybe one more magnitude compared to the length of the interval (0.1681 in this case)

Typo tinv(0.975,..)

Rules of thumb!

Example
$\bar{x} = 5.45$
$se(\bar{x}) = .041$

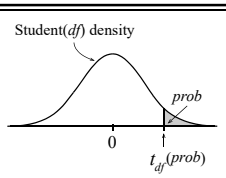
Typo ↷ read estimate instead of se"

If we calculate
CI = [23.76, 45.12]
order of magnitude is 1 ≈ 10
So present as [24, 25]

Typo: one more "order of"

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A 95% confidence interval



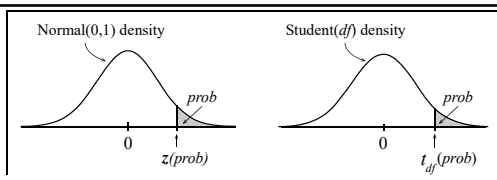
- When we estimate a population parameter θ using an estimate $\hat{\theta}$, the defining property of a 95% confidence interval is
  - "a type of interval that contains the true value of θ for 95% of samples"
- In our examples: a confidence interval is the $\hat{\theta} \pm t$ standard errors

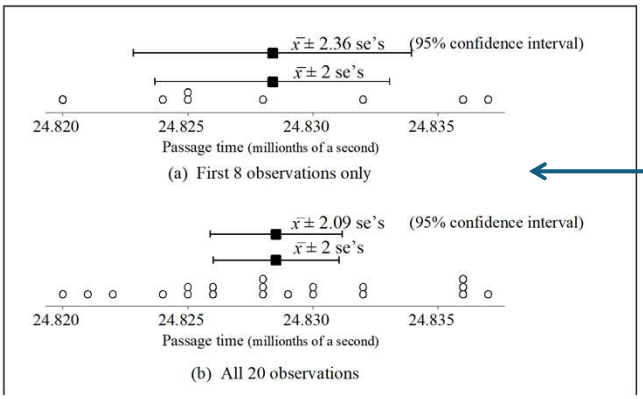
Value of the Multiplier, t, for a 95% CI

df :	7	8	9	10	11	12	13	14	15	16	17
t :	2.365	2.306	2.262	2.228	2.201	2.179	2.160	2.145	2.131	2.120	2.110
df :	18	19	20	25	30	35	40	45	50	60	∞
t :	2.101	2.093	2.086	2.060	2.042	2.030	2.021	2.014	2.009	2.000	1.960

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Review example





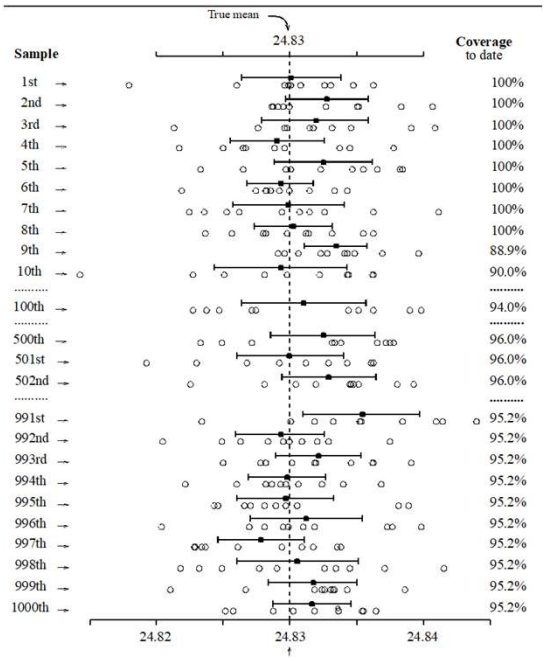
Newcomb's experiment to measure the speed of light in the year 1882

Time for a distance of 3721 m across the river Potomac

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Example

- 1000 experiments independent samples of size 10 from $\mathcal{N}(\mu = 24.83, \sigma^2 = 0.005^2)$
- In each case the 95% confidence interval is overlaid
- What is your observation?
- Repeated sampling
- Many independent studies
- Planning a study: In my study I will capture the correct value 95% of the time



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Review

- Effect of increasing the confidence level (df = ∞)

80% CI, $\bar{x} \pm 1.282\ \mathrm{se}(\bar{x})$

90% CI, $\bar{x} \pm 1.645\ \mathrm{se}(\bar{x})$

95% CI, $\bar{x} \pm 1.960\ \mathrm{se}(\bar{x})$

99% CI, $\bar{x} \pm 2.576\ \mathrm{se}(\bar{x})$

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Effect of increasing sample size

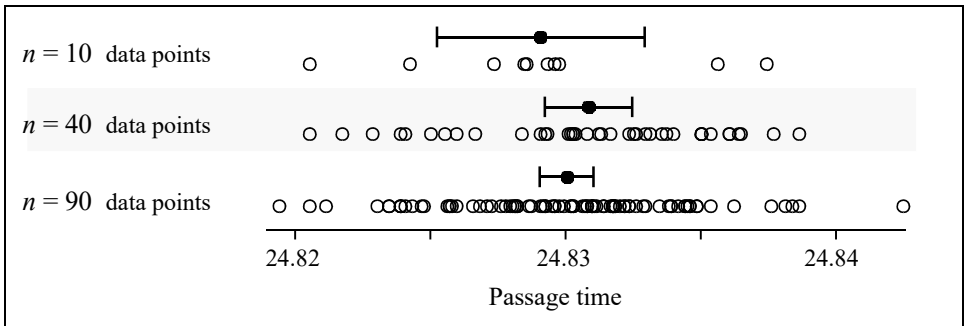
- If we have n observations and we want to increase the precision (say double it), how many more observations do we need?

menti.com
3350 7703

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Effect of increasing sample size

- If we have n observations and we want to increase the precision (say double it), how many more observations do we need?



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But what exactly is this t-distribution?

I can visualize the plot, I get df, I can use it to get the confidence interval, I can do significance digits but …

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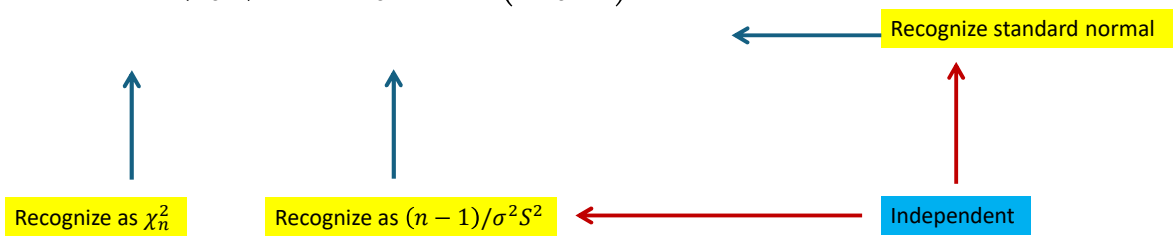
The t-distribution [1/2]

- Definition: If $Z \sim \mathcal{N}(0,1)$ and $V \sim \chi^2_{\mathrm{n}}$ are independent then
  - $T = \frac{Z}{\sqrt{V/\mathrm{n}}}$ is said to be a t-distribution with degrees of freedom *n*
- We have been using the result that if population were known to be normal, then $(\bar{X} - \mu)/(S/\sqrt{n}) \sim t_{n-1}$
  - We must therefore show that the LHS is a ratio of a standard normal to the square root of an appropriate term
  - Divide the numerator and denominator by $\sigma/\sqrt{n}$
    - To recognize that the numerator is standard normal
    - And denominator is S/σ that is S^2/σ^2

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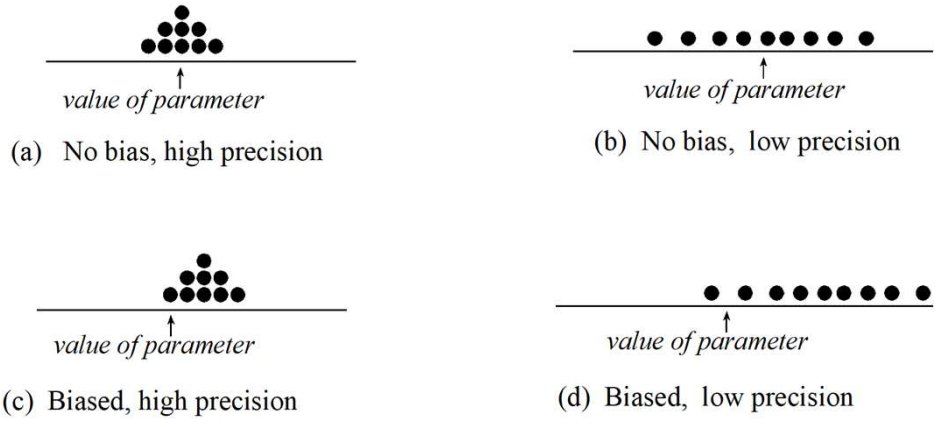
The t-distribution [2/2]

- But
  - $S^2 = \frac{\sum_{i=1}^{n}(X_i - \overline{X})^2}{n-1}$ and $\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}$
  - And recall the independence from below to complete the result
  - $\sum_{i=1}^{n}\left(\frac{X_i-\mu}{\sigma}\right)^2 = \frac{\sum_{i=1}^{n}(X_i-\overline{X})^2}{\sigma^2} + \left(\frac{\sqrt{n}(\overline{X}-\mu)}{\sigma}\right)^2$



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Bias and precision



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Unbiased Estimator

- The estimator $\bar{X}$ is an example of an unbiased estimator of the population mean because the expected value of $\bar{X}$ is μ
- The ***standard error*** of any estimate $\hat{\theta}$[denoted se($\hat{\theta}$)] estimates the variability of $\hat{\theta}$ values in repeated sampling and is a measure of the ***precision*** of $\hat{\theta}$

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